

BURGER: Benchmarking and Understanding Robotic Generalization via Embodied drawing

Abstract—How effectively can vision-language-action (VLA) models learn diverse language-conditioned behaviors as training data scales? Investigating this question requires large and varied robot datasets, whose collection on physical robots remains costly. We introduce BURGER, a language-conditioned robotic drawing benchmark with a scalable, prompt-driven pipeline for generating continuous drawing trajectories, both in simulation and the real world. In BURGER, an agent starts from the same blank canvas and controls a robot arm to draw the object or spatial composition specified by language, while each instruction allows an open-ended variety of valid drawings and stroke orders. Using 28,200 simulated demonstrations spanning 282 drawing tasks, we evaluate five representative VLA models across data scales, language grounding, and multimodal behavior. BURGER supports fine-grained analysis of language grounding, spatial composition, and action generation, revealing distinct failure patterns across VLA models. Our experiments suggest that some VLA models can benefit from greater task diversity and demonstration scale, capture multimodal drawing behaviors, and generalize to unseen object compositions of familiar objects. Finally, we deploy a trained policy on a Franka Panda arm, demonstrating the ability to draw reactively under human intervention. We refer the reader to <https://burger-bench.github.io> for more visualization.

I. INTRODUCTION

Vision-language-action (VLA) models integrate visual perception, language understanding, and action generation into policies for robotic control [1], [2], [3], [4]. As VLA models increasingly benefit from larger and more diverse datasets, an important question is whether current architectures are ready to effectively exploit large-scale language-conditioned robot trajectories. Investigating this question requires robot datasets with controllable scale and diversity, together with evaluations that distinguish different aspects of model behavior [5], [6], [7]. However, collecting diverse demonstrations on physical robots remains costly [8], [9], which motivates a more scalable setting for such studies.

Robotic drawing provides a complementary data regime with several practical advantages for such studies. Starting from the same blank canvas, the agent cannot infer the requested drawing from the initial observation alone, eliminating a potential visual shortcut [10]. Language prompts can be systematically increased in semantic complexity by composing objects, attributes, and spatial relations. The same instruction can admit many valid trajectories that differ in stroke order, starting point, and drawing direction, this intrinsic multimodality forces model to understand the concept, rather than memorizing the action sequence. The resulting canvases and execution trajectories make semantic, spatial, and action-execution failures directly inspectable, supporting

fine-grained diagnosis of model behavior. Drawing therefore provides a scalable setting for testing whether VLA architectures can learn complex language-conditioned and multimodal robot trajectories before scaling up costly real-world data collection.

Motivated by these properties, we introduce BURGER, a scalable benchmark and data-generation pipeline for language-conditioned robotic drawing, as illustrated in Fig. 1. Starting from a natural-language prompt, our pipeline generates diverse target sketches and converts them into feasible robot end-effector trajectories that can be executed in simulation and on a real robot. Using this pipeline, we construct a dataset containing 132 single-object tasks and 150 compositional tasks, with 100 demonstrations per task. The resulting data covers substantial variation in object semantics, spatial composition, and trajectory realization. As shown in Figure 2, BURGER exhibits substantially richer within-instruction planar trajectory diversity than LIBERO, while capable VLA policies can preserve multiple modes of this distribution after training. This makes robotic drawing a naturally multimodal data regime for studying VLA behavior. We further introduce a DINOv3-based evaluation metric that correlates strongly with human judgments, enabling scalable quantitative evaluation of generated drawings.

We use BURGER to evaluate several representative VLA models and study their ability to learn from robotic drawing trajectories. Our experiments reveal three main observations. First, VLA performance improves as either the number of demonstrations per task or the number of training tasks increases, suggesting that existing architectures can benefit from scaling robotic drawing data. Second, the π -family models can effectively fit the multimodal drawing trajectories and autonomously produce recognizable drawings after fine-tuning. Third, models exhibit non-trivial generalization beyond the training distribution, including generating unseen object compositions. At the same time, these evaluations expose characteristic failures in semantic grounding, spatial composition, and geometric precision that are difficult to distinguish through task-success metrics alone.

We do not intend robotic drawing to replace manipulation benchmarks or to evaluate the full spectrum of physical reasoning required for real-world robotics. Drawing largely abstracts away contact-rich interaction, object dynamics, and many sources of uncertainty that are central to manipulation. Instead, because drawings can be naturally represented as continuous robot trajectories, they provide a controlled setting for isolating several capabilities that are also important to VLA models: grounding increasingly complex language

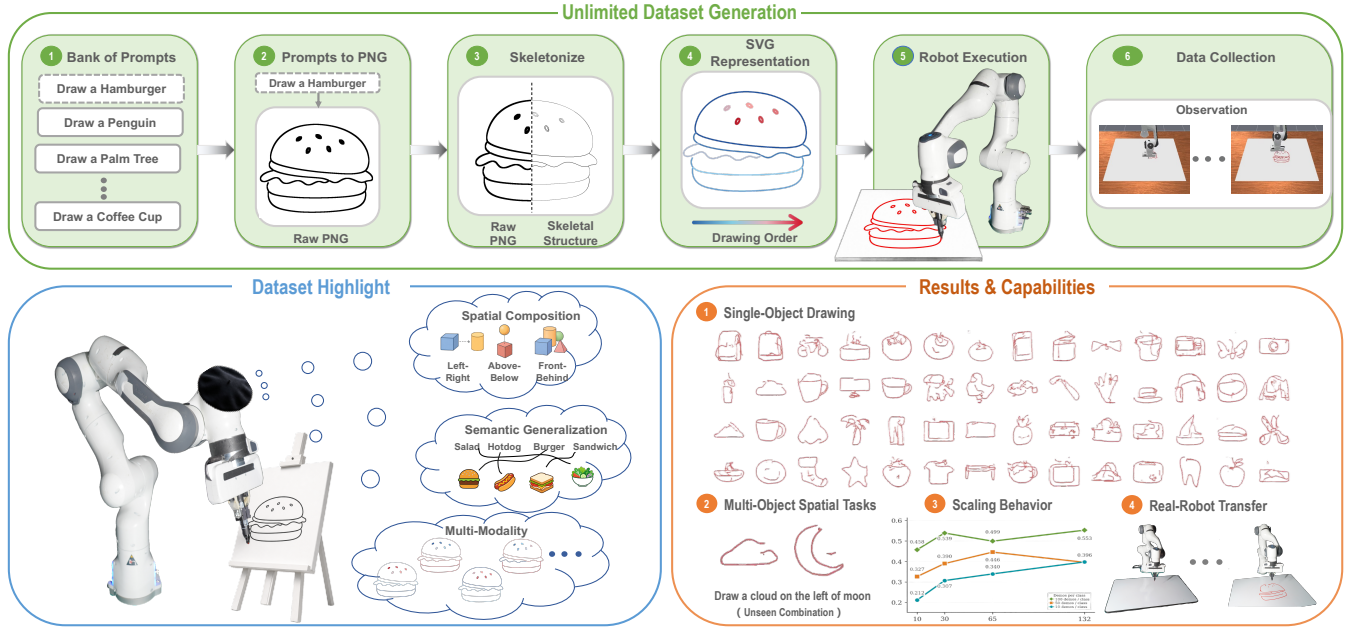


Fig. 1: Overview of BURGER. **Top:** Prompt-generated PNG images are converted into skeletons, SVG strokes, and robot trajectories. **Bottom left:** BURGER tests semantic grounding, spatial reasoning, and multimodal action generation in VLA models. **Bottom right:** We evaluate models on a wide range of aspects.

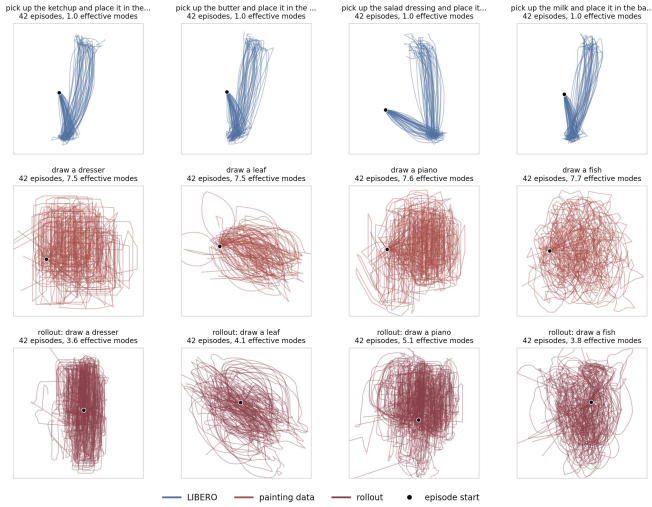


Fig. 2: Instruction-conditioned planar trajectory diversity. The three rows show LIBERO demonstrations, BURGER demonstrations, and $\pi_{0.5}$ rollouts, respectively, with trajectories overlaid for the same instruction. Different modes correspond to distinct clusters of execution routes.

instructions, generating long-horizon continuous actions, representing multimodal behaviors, and exploiting larger and more diverse datasets. In this sense, BURGER serves as a complementary diagnostic benchmark for studying the readiness and limitations of current VLA architectures.

II. RELATED WORK

Vision-language-action (VLA) models map visual observations and language instructions to robot actions. From the perspective of action generation, recent VLA methods can be roughly grouped into autoregressive and continuous

generative paradigms. Autoregressive VLAs, such as OpenVLA [2], decode actions through a sequential prediction process akin to language models. Recent works improve this paradigm with better action representations, action chunking, parallel decoding, and unified token sequences [2], [11], [12], [13], [14], [15]. Continuous generative VLAs instead model continuous action chunks using flow matching or diffusion [16], [17]. Models such as π_0 and $\pi_{0.5}$ use flow matching for continuous action generation [18], [19], while RDT-1B and GR00T N1 employ diffusion-based action generation [20], [21]. Recent hybrid methods, such as Diffusion-VLA [22], combine autoregressive reasoning with diffusion-based action generation. BURGER focuses on evaluating these action-generation paradigms under highly multimodal, long-horizon, and semantically rich drawing data.

The benchmark for VLA systems has moved beyond measuring binary task success rates toward more diagnostic evaluation. Recent benchmark studies VLA’s ability in long-horizon reasoning, spatial and semantic grounding, robustness to scene or language perturbations, structured behavioral metrics, and memory-dependent manipulation [6], [7], [23], [24], [25]. These efforts show that similar success rates can hide different failure modes in grounding, planning, robustness, or temporal consistency. BURGER follows this capability-oriented evaluation direction, but uses language-conditioned drawing as the diagnostic domain. Unlike goal-oriented manipulation, drawing admits many valid stroke orders, geometries, and trajectories for the same prompt. This makes the more dedicated abilities directly observable from the generated canvas and rollout trajectory.

Data scaling in VLA learning has been studied in robot imitation learning and generalist policy training. Prior

work analyzes how data quantity and data composition affect manipulation-policy generalization, including object and environment diversity, camera viewpoints, spatial arrangements, and dataset fragmentation [26], [27], [10]. BURGER considers a related scaling question in a controlled language-conditioned drawing domain.

Robotic drawing and painting have been studied primarily as a control and planning problem. For example, Lee et al. [28] formulate robotic sketching as a deep decoupled hierarchical reinforcement learning problem, where separate policies learn stroke-based rendering and motor control from scratch. FRIDA [29] develops a collaborative robot painting system with a differentiable real2sim2real planning environment, enabling a robot to generate paintings from high-level human inputs such as language descriptions or reference images. More recently, IMPASTO [30] studies robotic oil-painting reproduction by integrating learned pixel dynamics models with model-based planning. These works advance robotic drawing by improving stroke planning, motor control, and real-world execution. In contrast, our work studies sketching as a language-conditioned VLA benchmark.

III. METHOD

A. Task Formulation

At time t , a policy receives a language instruction ℓ , a canvas observation I_t , robot proprioception q_t , and predicts an end-effector command a_t (or an action chunk, depending on the model). The task is to produce the requested object or spatial composition through sequential drawing actions. The policy is not given a target sketch at inference. Canvas observations change as strokes accumulate, while pen-lifting transitions separate disconnected strokes. The benchmark therefore involves both drawing geometry and decisions about when to move without marking the canvas.

B. Prompt-to-Trajectory Data Generation

Figure 1 summarizes the pipeline. **PNG Generation.** Diffusion-based image generation models have shown strong capabilities in producing diverse visual content from text prompts [31]. We use GPT-Image 2 [32] to generate the initial sketch-style PNG images from the corresponding language prompts.

PNG to Skeleton. The PNG images are binarized into black and white and converted into skeletons using threshold-based foreground extraction, connected-component analysis, and morphological skeletonization algorithm. This step also extracts the contours of solid black regions like eyes, creating lines suitable for robot drawing.

Skeleton to SVG. We decompose the skeleton into individual strokes by detecting connected components and order them using a weighted heuristic that encourages drawing from top to bottom, outlines then details, and favoring transitions to nearby strokes. We visualize the generated SVG image in Figure 1, showing stroke sequence via color brightness.

SVG to Trajectory. We use ManiSkill [33], a robotic manipulation simulation platform, to replay each SVG with

a robot arm. The simulator includes a built-in drawing environment where visual marks appear on a canvas when the end-effector comes close to its surface. We use an end-effector pose controller to follow the strokes in sequence and insert pen-lifting transitions between disconnected strokes. During execution, we record RGB observations, robot proprioception, and actions as (observation, action) pairs. The data is stored in HDF5 format and then converted to the corresponding robot-dataset format for training. Although our main experiments use ManiSkill with the SAPIEN backend, the trajectory-generation interface is not tied to a specific simulator: we additionally implement the same pipeline in other simulation platforms, as shown in Figure 4.

C. The BURGER Dataset

To collect representative and diverse drawing trajectories, we adopt the setting from the Google Quick, Draw! [34] dataset, which contains 345 object classes and millions of human-drawn sketches. We select 132 classes that are commonly found in the real world and can be easily drawn by a robot arm. We further define 150 compositional classes by pairing two of the 100 base classes with spatial relations. Among them, 100 classes use left-right relations, while the remaining 50 use vertical relations such as “over” and “under”. For each class, we generate 100 trajectories using the pipeline described above, resulting in 15,000 compositional trajectories and 28,200 trajectories in total.

To quantitatively characterize the advantage of BURGER in providing multimodal trajectory data, we compare its instruction-conditioned trajectory diversity with all four LIBERO suites, as visualized in Figure 2. Since BURGER focuses on drawing rather than full-space manipulation, we compare planar end-effector trajectories on the drawing plane. We use a count-matched setting of 40 instructions and approximately 42 episodes per instruction. Each trajectory is resampled to 32 points and represented by its start-relative xy path and step vectors. A shared k -means codebook is fitted on the pooled trajectory descriptors, and the effective number of trajectory modes is defined as

$$N_{\text{eff}} = \exp \left(- \sum_{j=1}^k p_j \log p_j \right), \quad (1)$$

where p_j denotes the fraction of trajectories assigned to cluster j . We use $k = 24$ for the reported results and observe consistently higher multimodality for BURGER across different choices of k . Under this setting, LIBERO contains 1.09 effective modes per instruction on average, compared with 7.13 for BURGER, corresponding to a $6.6\times$ increase. The same trajectory representation and clustering procedure is also used to analyze policy rollouts in Sec. IV-C.

We also collect real-robot trajectories to evaluate whether the BURGER data pipeline can support real-world policy learning. The real-robot dataset follows the same data format as the simulated dataset and includes two RealSense camera views from the front and back of the workspace; either view can be used as the visual observation input. Figure 3

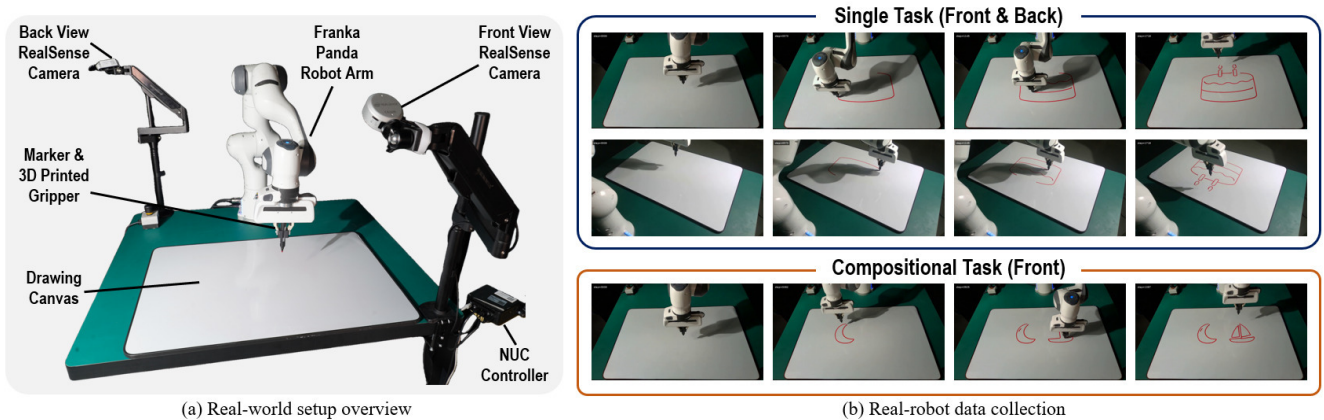


Fig. 3: Real-world deployment of BURGER. **Left:** The Franka Panda hardware setup equipped with a drawing tool. **Right:** The real-robot data collection process for a single task “birthday cake” and a compositional task “moon left sailboat”.

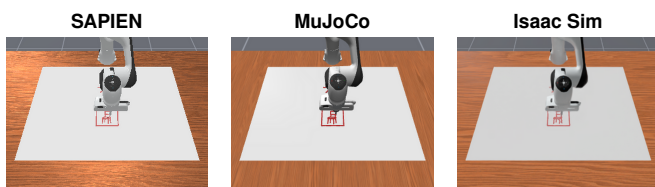


Fig. 4: Cross-simulator deployment of the BURGER trajectory-generation pipeline. The same SVG drawing task is executed in SAPIEN, MuJoCo, and Isaac Sim using corresponding robot-control interfaces.

shows the hardware setup and real-robot data-collection process. The same pipeline supports both single-object and compositional drawing tasks, as illustrated by the “birthday cake” and “moon left of sailboat” examples. In this work, we collect 10 single-object classes with 10 demonstrations per class for real-robot training and evaluation.

The dataset is, to our knowledge, the first large-scale VLA dataset for language-conditioned drawing tasks, and we hope it can serve as a valuable resource for future research on VLA model generalization.

D. Quantifying Robot Trajectory Quality

Faithfully evaluating drawing quality is challenging because sketch generation is inherently multimodal: the same object can be rendered with different stroke orders, shapes, and local geometric variations. Contrastive metrics such as CLIPScore [35] measure global image–text alignment, but prior work [36] shows that such metrics can fail to detect fine-grained errors. For example, a metric may assign a high score to a drawing of “an apple to the right of a pear” even when the prompt specifies the opposite. To obtain a more human-aligned and fine-grained evaluation metric, we design a contrastive feature-based evaluator trained directly on the BURGER drawing distribution.

Our evaluator builds a sketch-specific metric space on top of a frozen DINOv3 backbone [37]. We combine the CLS token and mean-pooled patch embeddings and map them through a lightweight two-layer MLP into a normalized sketch embedding. The projection head is trained with a co-

TABLE I: Agreement with human judgments.

Metric	Spearman ρ	Kendall τ_b	Pairwise Agreement
CLIPScore [35]	0.592	0.433	0.803
Ours	0.834	0.655	0.875

sine classifier using classification and supervised contrastive objectives, encouraging drawings of the same object class to form compact clusters. After training, each object class is represented by the prototype of its training embeddings, and generated drawings are scored by their similarity to the target-class prototype. For compositional tasks, we locate the minimum-connectivity boundary between the two foreground regions to separate the objects, allowing us to evaluate each object individually and determine whether their spatial relation matches the instruction.

We validate the metric through a user study covering 10 single-object classes. For each class, 66 participants rank five drawings, including four generated by VLA policies and one reference drawing, according to how well they depict the prompted object. We compare the resulting human judgments with our metric and CLIPScore [35] using Spearman correlation, Kendall’s τ_b , and pairwise agreement. As shown in Table I, our metric achieves substantially stronger agreement with human judgments across all three measures, supporting its use as a scalable evaluation metric for language-conditioned drawing policies.

IV. EXPERIMENTAL RESULTS

Our experiments evaluate the capabilities and limitations of VLA models on language-conditioned robotic drawing tasks in BURGER. Concretely, we investigate three questions: (1) whether BURGER can serve as a diagnostic benchmark for comparing VLA models across several dimensions, including data scaling, language grounding, spatial-relation understanding, and multimodal behavior modeling; (2) whether the benchmark can expose characteristic failure behaviors and help diagnose differences among VLA formulations; and (3) whether policies trained with BURGER data can transfer to real-world drawing and support closed-loop interaction under human modifications.

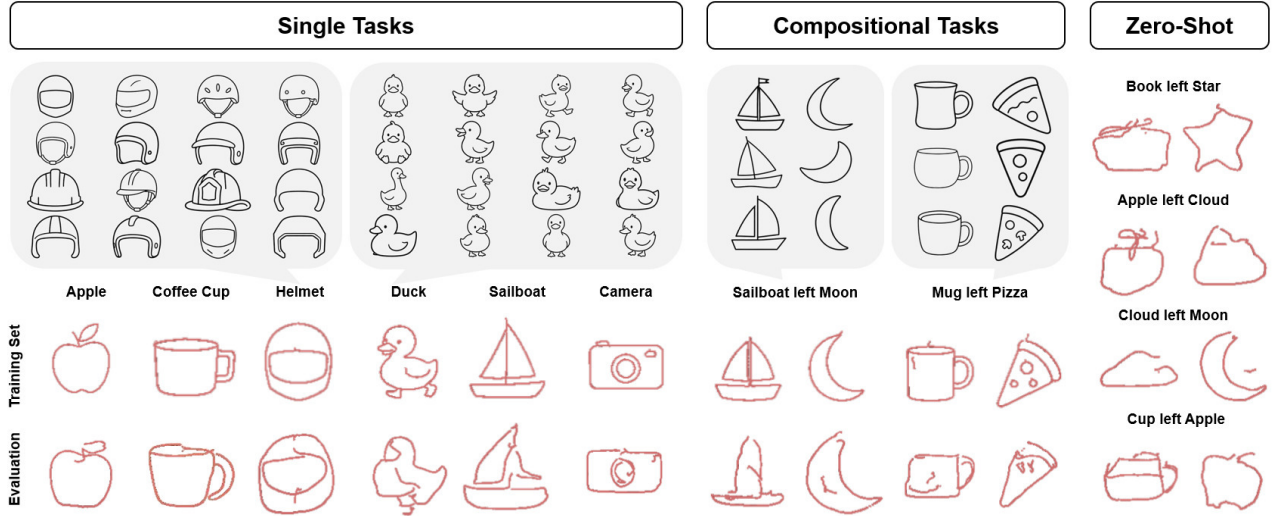


Fig. 5: Dataset diversity and VLA evaluation in BURGER. We show representative single-object and compositional training tasks, along with corresponding VLA rollouts. The rightmost column evaluates generalization to unseen spatial compositions.

Baselines. We evaluate representative VLA models with different action-generation formulations. π_0 [3] uses a separate action expert to generate continuous actions using flow matching. $\pi_{0.5}$ [4] extends π_0 with co-training on heterogeneous tasks for broader generalization. π_0 -FAST is an autoregressive VLA based on the FAST action tokenizer [12]. **OpenVLA** [2] is an open-source autoregressive VLA model. **OpenVLA-OFT** [11] is an optimized fine-tuning variant of OpenVLA that predicts continuous action chunks with parallel decoding rather than step-wise discrete action tokens.

A. Scaling Behavior of VLA Models

We evaluate the OpenPI-family VLA models across different task scales, varying the number of object classes from $C = 10$ to $C = 132$ and the number of trajectories per class from $T = 10$ to $T = 100$, as shown in Figure 6. Drawing performance is evaluated using the DINOv3-based target similarity metric introduced in Sec. III-D. Overall, increasing either the number of training classes or the number of trajectories per class generally improves drawing performance, particularly for π_0 and $\pi_{0.5}$. However, the scaling behavior is not uniform across models: π_0 -FAST shows

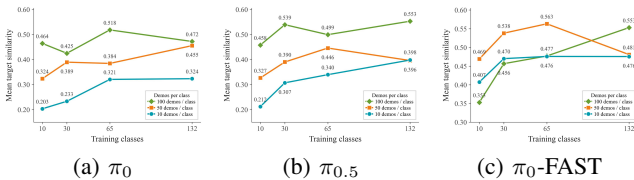


Fig. 6: Scaling behavior of π_0 , $\pi_{0.5}$, and π_0 -FAST under different numbers of training classes and demonstrations per class. The flow-matching models generally benefit from increasing data scale, while π_0 -FAST shows more variable performance and appears more sensitive to the multimodal data distribution.

larger fluctuations and does not consistently benefit from increasing data scale. These results suggest that BURGER provides a controlled setting for studying how different VLA formulations respond to increasing task diversity and demonstration scale.

B. Language-conditioned Semantic and Spatial Generalization

We diagnose language grounding and semantic generalization across the OpenPI-family models from two dimensions. First, we evaluate language-conditioned instruction following by varying the object names in the prompts. Starting from the same blank canvas, different language instructions lead the policies to generate different objects corresponding to the requested categories. Across the in-distribution tasks, we observe no evident confusion between object instructions. In contrast, when the instruction specifies an object category that is not included in the training dataset, the models fail to generate the corresponding object and instead tend to produce similar, unstructured strokes across different unseen instructions. This suggests that the learned language grounding is effective for the trained semantic categories but does not directly generalize to unseen object concepts.

Second, we evaluate spatial understanding using compositional drawing tasks with prompts of the form “draw A on the left of B”, “draw A under B”, and “draw A over B”. Our goal is to study whether VLA models can generalize simple spatial relations beyond memorized object compositions. For this evaluation, policies are trained on these compositional tasks. As shown in the right-most column in Figure 5, VLA models can generate valid left-right and over-under object compositions even for object pairs that are not observed during training. For example, a policy can draw a cup to the left of an apple despite never being trained on this specific composition. We also observe that the policies correctly distinguish different spatial instructions. These results suggest

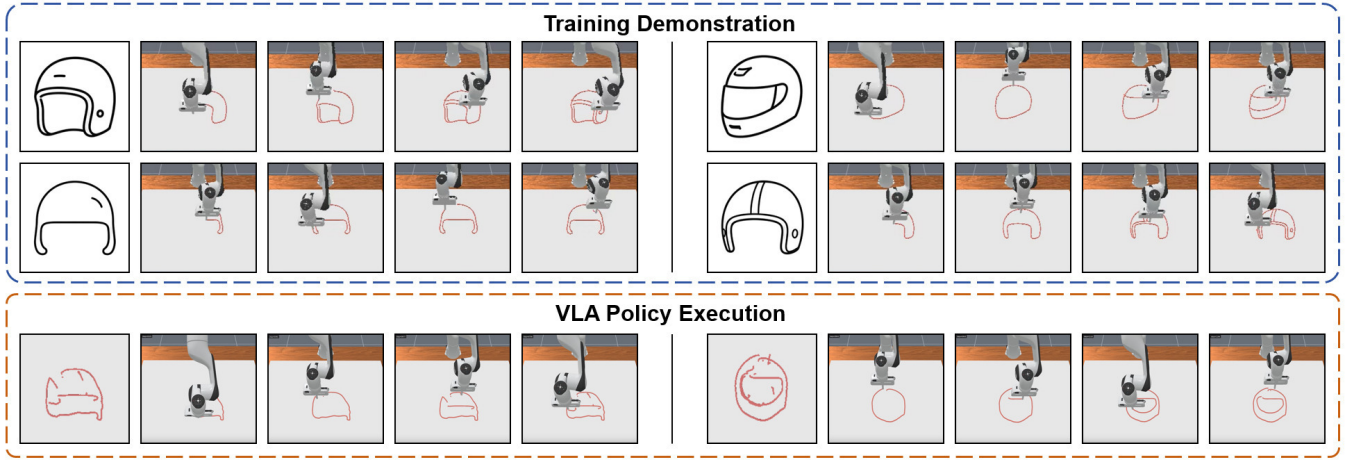


Fig. 7: Learning multimodal drawing behaviors in BURGER. **Top:** different drawing patterns of “helmet” within the dataset, **Bottom:** the generated images by $\pi_{0.5}$ unseen in the dataset.

that the policies can generalize beyond memorized object compositions and apply learned spatial relations to unseen object pairs.

C. VLA Models Capture Multimodal Drawing Behaviors

Sketch generation naturally admits multimodal trajectories for the same language instruction. As quantified in Sec. III-C and visualized in Figure 2, BURGER exhibits substantially stronger within-instruction multimodality than LIBERO. We therefore use it to evaluate whether VLA policies can learn and preserve such multimodality after training.

We first evaluate this quantitatively by applying the same trajectory representation and clustering procedure to $\pi_{0.5}$ rollouts. As shown in Figure 2, the training data for the four representative tasks contains approximately 7.5 effective trajectory modes per instruction, while the learned policy retains 3.6–5.1 modes across the corresponding rollouts. These results show that $\pi_{0.5}$ can preserve multimodality in its rollout distribution after training on BURGER.

We further examine this behavior qualitatively in Figure 7. The first two rows show distinct demonstrations of the same object class, *helmet*, with variations in geometry, stroke ordering, and execution path. After training on these demonstrations, $\pi_{0.5}$ produces multiple valid helmet drawings under the same instruction. Together, the trajectory-level and visual results show that BURGER supports learning a distribution of valid drawing behaviors rather than a single deterministic route.

D. BURGER Reveals Fine-Grained VLA Failure Modes

Different VLA models exhibit distinct failure patterns, which BURGER can expose through process-level analysis of generated trajectories and drawings.

Single-Object Tasks. OpenVLA and π_0 -FAST are sensitive to the initial end-effector position: OpenVLA may become trapped in a local region, while π_0 -FAST can drift away from the target area. In contrast, flow-matching policies sometimes fail to terminate after completing the object, leading to overpainting.

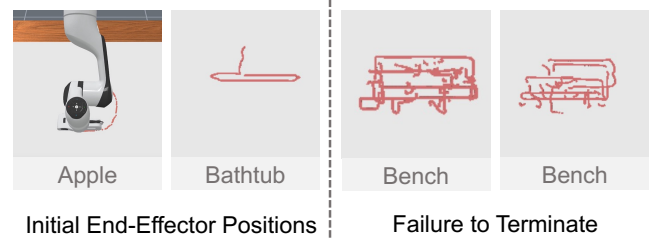


Fig. 8: Failure modes of VLA rollouts in single-object drawing tasks.

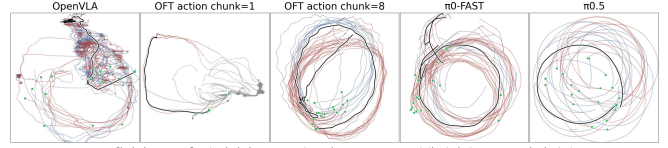


Fig. 9: Controlled circle-drawing results across representative VLA models, revealing distinct failure patterns in contact, trajectory progress, and multimodal behavior.

Compositional Tasks. Object quality may degrade under compositional instructions even when the spatial relation is correct. This degradation becomes more evident when the instruction space expands from horizontal to both horizontal and vertical relations, suggesting that relational grounding and object-level geometry become harder to preserve jointly.

To further diagnose these differences, we construct a controlled circle-drawing case study with 100 demonstrations sharing the same shape and radius, while varying the circle center, starting point, and drawing direction. OpenVLA often maintains contact but loses planar progress, whereas OpenVLA-OFT with chunk length 1 frequently fails to establish stable contact. Within OpenVLA-OFT, increasing the chunk length from 1 to 8 substantially improves contact and trajectory coherence, indicating a clear effect of action chunking on long-horizon execution.

However, coherent execution does not guarantee multimodal coverage. OpenVLA-OFT with action chunks and π_0 -

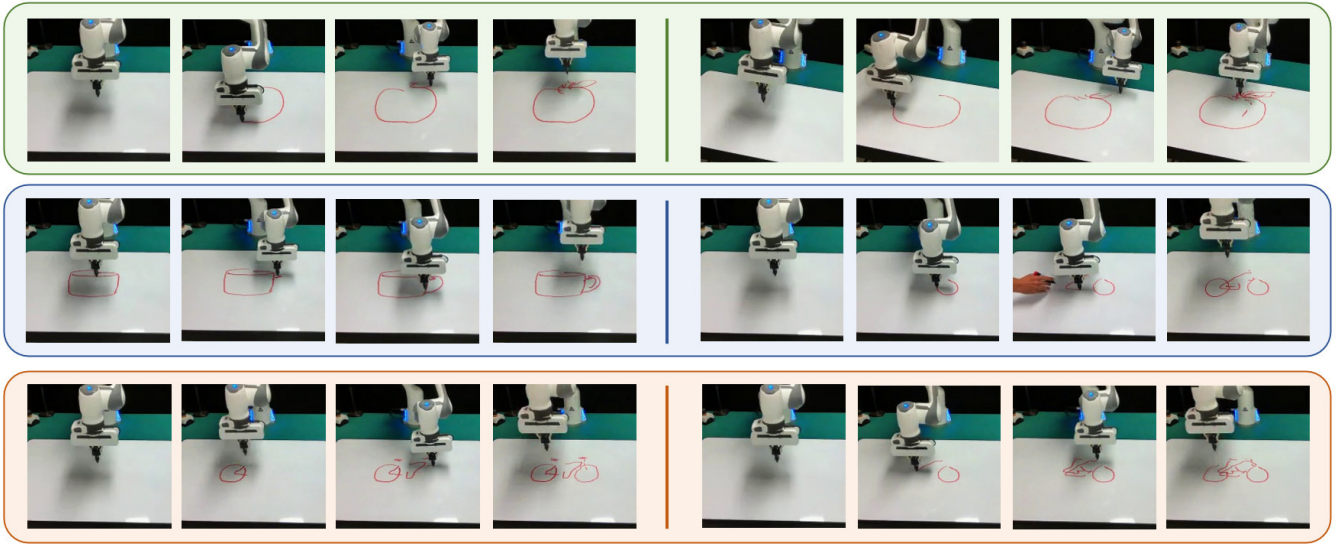


Fig. 10: Real-robot drawing results. The first row shows robustness to camera-angle changes; the second shows visual feedback through object completion and human intervention; the third shows multimodal real-robot behaviors.

FAST tend to concentrate on a subset of starting points and drawing directions, while the flow-matching policies π_0 and $\pi_{0.5}$ better preserve trajectory diversity. These results suggest that action chunking mainly affects execution coherence, whereas the action-generation formulation also influences multimodal mode coverage. This case study illustrates how the BURGER pipeline can support controlled diagnosis of model-specific failure patterns.

E. Real Robot Results

Using the real-robot demonstrations introduced in Sec. III-C, we fine-tune $\pi_{0.5}$ directly on data collected with the Franka Panda platform. As shown in Figure 10, the resulting policy can reliably execute real-world drawing tasks, demonstrating that the BURGER data-collection pipeline can directly support real-world policy learning. The first row further shows that the policy remains effective under changes in camera viewpoint, indicating moderate robustness to visual observation variations.

We further examine closed-loop visual feedback through completion and interaction tasks. Given a partially completed cup, the policy can add the missing handle. In the bicycle example, after a human draws the rear wheel, the robot observes the updated canvas, skips the completed component, and continues with the remaining structure. The third row additionally shows that the policy can produce different valid trajectories for the same drawing task, preserving multimodal behavior on the real robot. Together, these results demonstrate visual closed-loop interaction and multimodal real-world execution.

V. CONCLUSION AND LIMITATIONS

We present BURGER, a language-conditioned robotic sketch-painting benchmark to evaluate VLA models through continuous stroke generation. It enables unlimited robotic

data generation with rich data diversity, fine-grained evaluation of semantic grounding, spatial reasoning, multimodal behavior modeling, and action generation beyond binary task success rates. Real-robot experiments further demonstrate the feasibility of transferring trained policies to sketch drawing.

Although BURGER provides a scalable and diagnostic testbed for evaluating language-conditioned robotic drawing, several limitations remain in our benchmark. First, our setup abstracts away many physical complexities of painting. Unlike robotic painting systems that model brush dynamics [30], [29], BURGER focuses on sketch-level stroke generation with a drawing tool. This simplification makes the benchmark scalable, but limits its coverage of contact-rich artistic media such as brush painting. Second, the dataset inherits biases from the automatic generation pipeline. Our stroke-order heuristics may introduce drawing styles that differ from human demonstrations. As a result, dataset may not fully represent natural human drawing behaviors.

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